



Module 21 Day 3: Autopsy and iPhone Forensics

Overview

Today, you will use Autopsy to analyze an iPhone image.

Class Objectives

By the end of the lesson, you will be able to:

- Identify the methods used in smartphone forensics investigations.
- Navigate the database and file structure of the iPhone's flash drive.
- Locate identifiable evidence on the iPhone to establish ownership.
- Use Autopsy to view and tag evidence in an iPhone image.
- Extract image content for offline viewing in other applications (logs, text, pictures, video, audio).

Lab Environment

Lab Details

In this module, you will continue to use the Forensics lab environment.

Username - `kali` Password - `kali`

- Slides for today's lesson can be found on Google Drive here: [21.3 Slides](#).

01. Welcome and Overview

In the last session, we covered:

- Basic principles of digital forensics methodologies

- Various skill sets required for a career in digital forensics
- The proper approach to collecting, preserving, analyzing, and reporting forensic evidence
- How to conduct a preliminary review for a forensic case
- How to preserve and document evidence using Autopsy

Today, we'll focus on smartphone forensics and continue the Pure Gold Credit Union case by using Autopsy to view and tag evidence on an iPhone image.

Where's the Data? File System and Data Storage

An important aspect of mobile forensics is understanding where data is stored, how to access the data, and what data can be recovered.

- An iPhone does not have external storage. All data is stored in flash memory on the device.
 - Remember: Flash memory retains data without any external power and is used in SSDs and smartphones.
 - Flash memory on an iPhone exists until it is overwritten.
- Data is first imaged using a bit-level copy.
 - In an iPhone, this will recover deleted messages, GPS coordinates, and cell tower locations.
- iPhones allow users to back up their data to the cloud. This can be another source of data in an investigation, especially when the device is not available.
 - What challenges does iCloud data present to a forensic investigator?
 - Answer: Isolating and securing the evidence is challenging when data is in multiple locations. Also, SLAs must be recognized when dealing with other companies, in addition to other legal issues.

Important Directories/Databases and Files

Now we'll look at directories, databases, and files in the iPhone image that you will investigate for evidence using Autopsy for the Pure Gold Credit Union case.

Since iPhone's iOS is based on Unix, the directory structure should be familiar.

- Most information is in human-readable format within the Autopsy application.
- The iPhone stores user data in SQL databases and other files.
 - SQL stands for Structured Query Language and is a language used to create, read, update, and delete (commonly abbreviated as "CRUD") database files.

- These are the main databases that applications such as mail, SMS, calendar, and the address book use (*Note: Your iPhone image has been reduced in size for this class, and will not contain all of the following files*)
 - AddressBook.sqlitedb contains contact information and personal data like name, email address, etc.
 - AddressBookImages.sqlitedb contains images associated with saved contacts.
 - Calendar.sqlitedb contains calendar details and events information.
 - Call_History.db contains incoming and outgoing call logs, including phone numbers and time stamps.
 - sms.db contains text and multimedia messages along with their time stamps.
 - voicemail.db contains voicemail messages.
 - Safari/Bookmarks.db contains saved URL addresses.
 - locationd contains Geolocation details.
 - Safari/History.db contains internet browsing history.

iPhones also have data stored in `plist`s, or property lists. A `plist` typically stores configuration information.

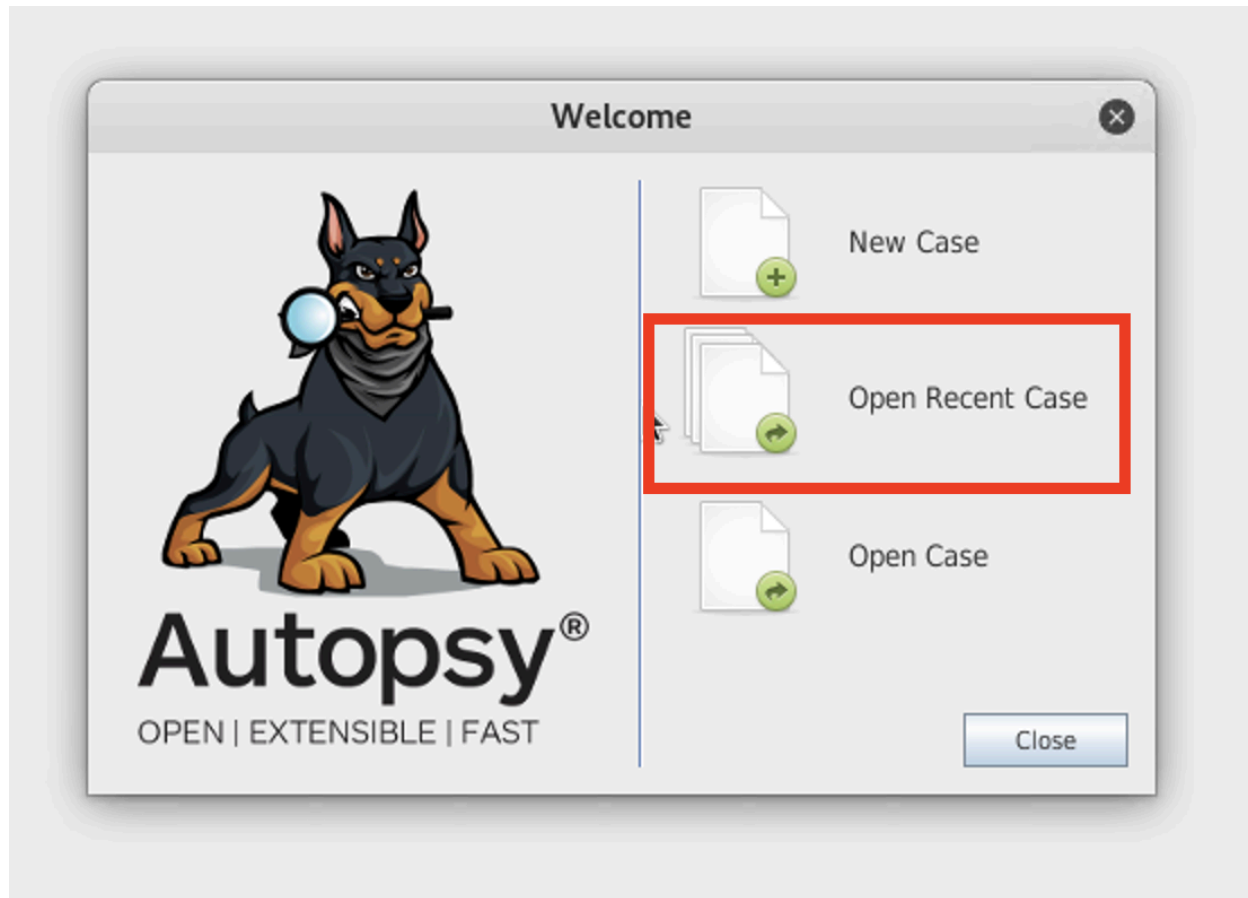
- `activation_record.plist` Model information - ICCID, IMEI, Serial Number
- `data_ark.plist` File associated with backup and restore, contains assortment of information including host name and OS version
- `Maps/Bookmarks.plist` contains maps and bookmarks.

Demonstration: Evidence Analysis with Autopsy

In this demo, we'll walk through various methods for obtaining digital evidence related to the Pure Gold Credit Union case:

1. Launch Autopsy:
 - Double-click on the Autopsy Icon on your desktop
2. Open the case created in the last class:

- Case name: Pure-Gold-Credit-Union



- The Autopsy files end with the .aut file extension.



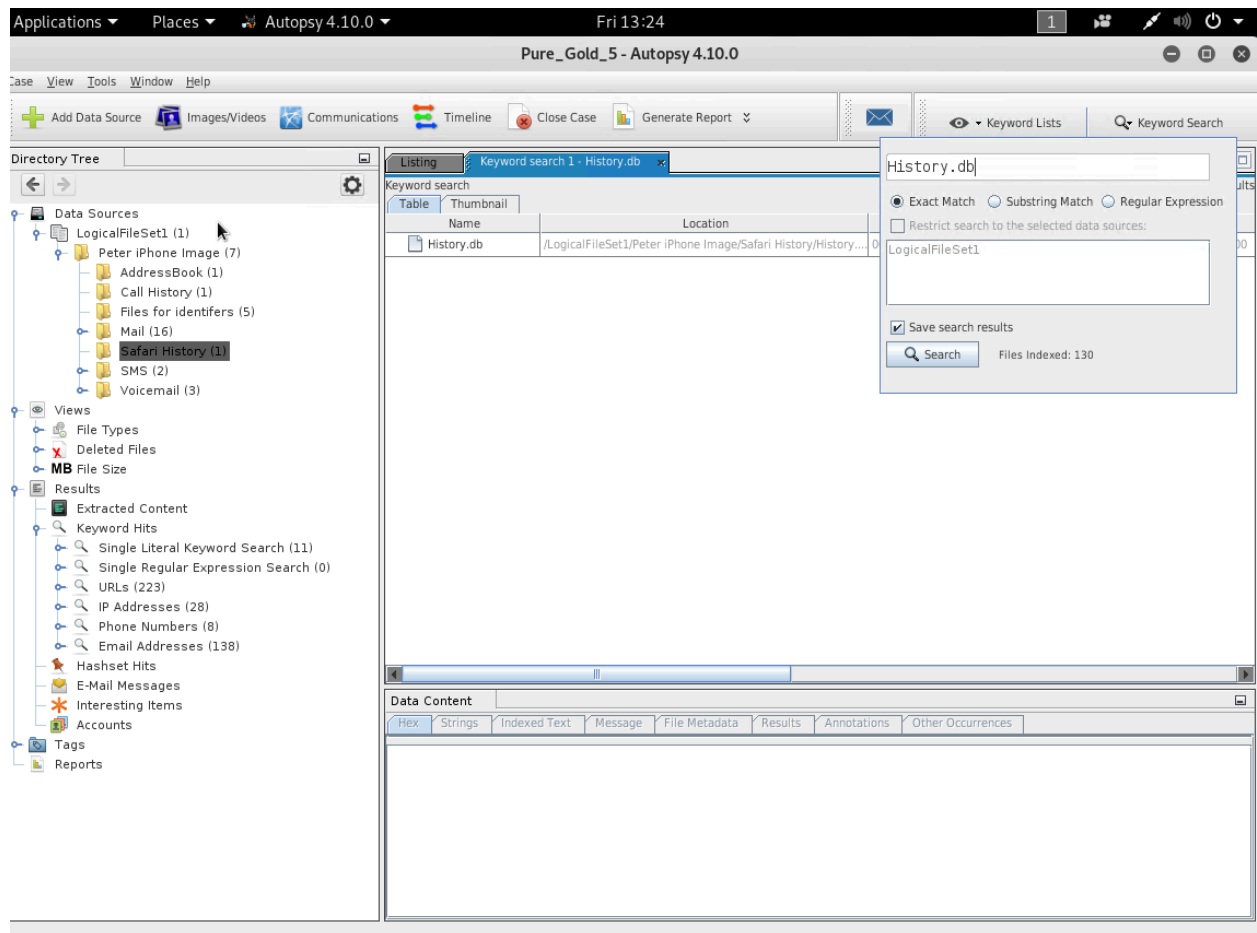
Examine Browsing History

This example shows how to navigate an evidence file without using the directory tree.

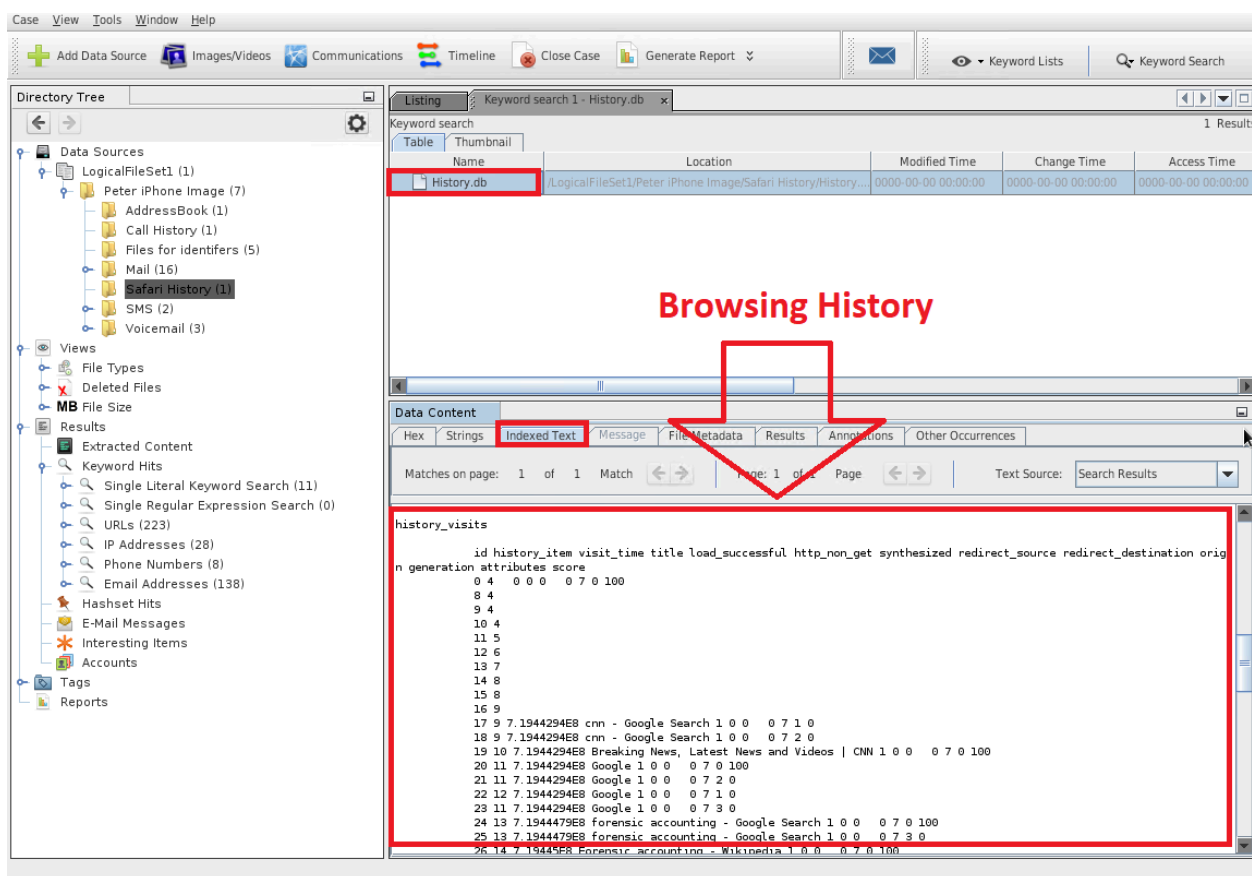
Clicking through directory trees is a long and cumbersome process. Instead, Autopsy has a built-in feature that allows for quicker and more thorough searches.

You will need the file name or file extension to perform a search.

1. From the Keyword Search menu, select Exact Match.
2. Type History.db.
3. Click the Search button.
4. Select History.db at the top of the File Search Results 1 window.
 - The History.db stores iPhone browsing history from the Safari web browser.



5. Select the Indexed Text tab in the Data Content pane.
 - Here you will see a list of searches that the owner of the iPhone performed.

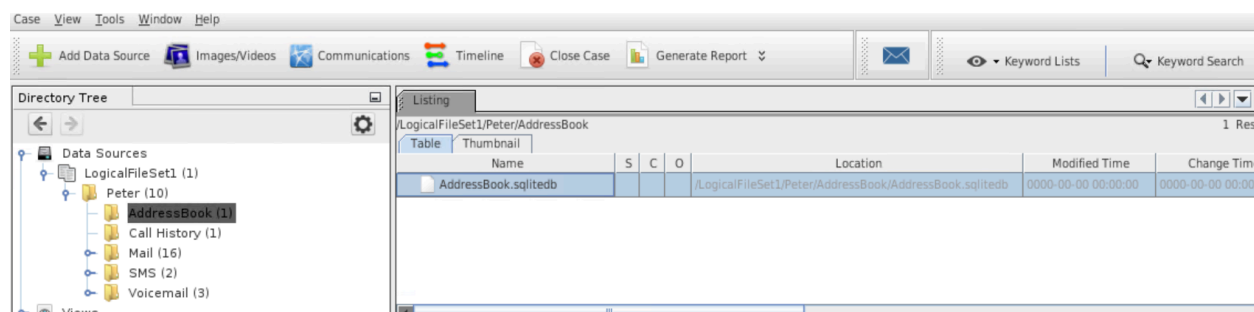


6.

Note how to view the text using the iPhone image file.

Autopsy Main Panes

- Review the main panes in the user interface: Directory Tree, Listing, and Data Content.



File Metadata

Next, we'll look at the file metadata. We need this information for our evidence reporting.

- Select the File Metadata tab in the Data Content pane.
- Note the importance of capturing file metadata using the File Metadata tab in the Data Content pane. Here we can find lots of information relevant to our investigation, including:
 - Image name
 - MIME type
 - File size
 - Hash

The screenshot shows a file analysis interface. At the top, a 'Listing' pane displays a table with file details. Below it, the 'Data Content' pane is active, showing the 'File Metadata' tab. The metadata for 'AddressBook.sqlitedb' is displayed in a key-value format.

Name	S	C	O	Location	Modified Time	Change Time
AddressBook.sqlitedb				/LogicalFileSet1/Peter/AddressBook/AddressBook.sqlitedb	0000-00-00 00:00:00	0000-00-00 00:00:00

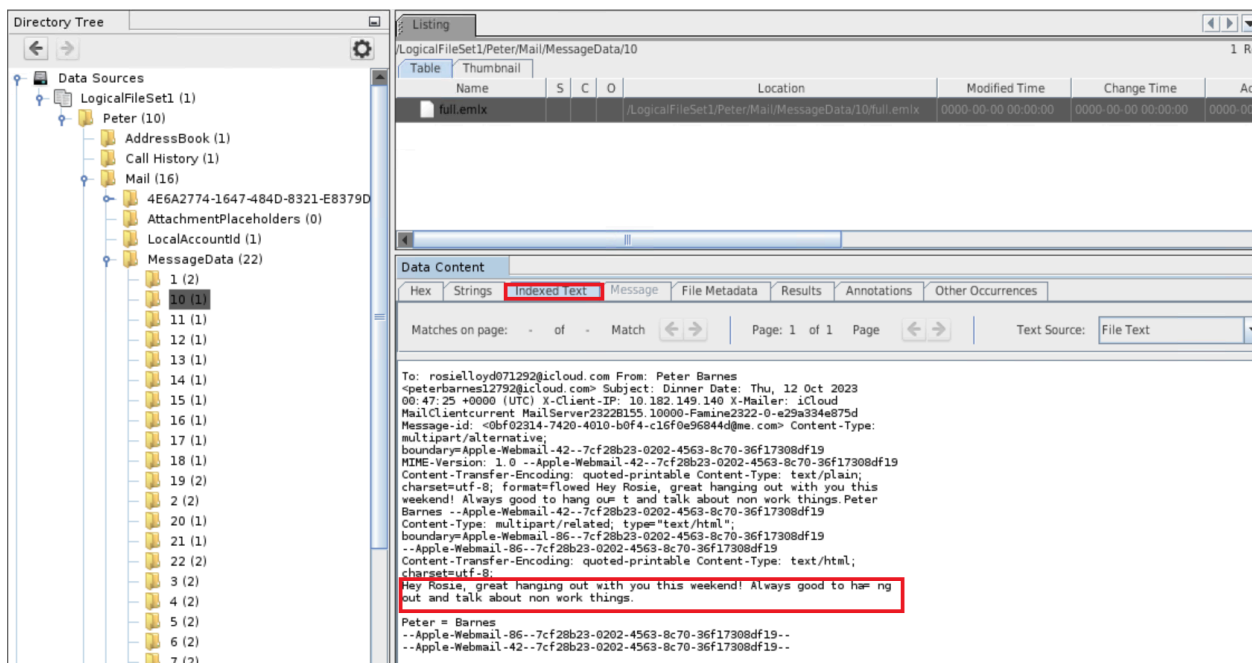
Data Content	
	File Metadata
Name	/LogicalFileSet1/Peter/AddressBook/AddressBook.sqlitedb
Type	Local
MIME Type	application/x-sqlite3
Size	307200
File Name Allocation	Allocated
Metadata Allocation	Allocated
Modified	0000-00-00 00:00:00
Accessed	0000-00-00 00:00:00
Created	0000-00-00 00:00:00
Changed	0000-00-00 00:00:00
MD5	1a75790ce6c0cad05c6da821e7399532

Displaying Text

The Indexed Text tab is used to display human-readable text.

- This example displays the contents of one of Peter's emails.

- The path to the file is displayed in the Listing pane.



File Search

- If you know the file name you're looking for, performing searches using the File Search by Attributes method is an alternative to using the directory tree.
 - To get there, click Tools in the top menu bar and select File Search by Attributes.
 - Next to the Keyword Search, the File Search by Attributes function is a highly efficient method for searching through file images.
 - *Note that this may take a few minutes to return the search results.*

Listing File Search Results 1 File Search Results 2

Filename Search Results: 1

Name	S	C	O	Location	Modified Time	Change Time	Access Time
sms.db				/LogicalFileSet1/Peter/SMS/sms.db	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00

File Search by Attributes

☒ Name:

*Note: Name match is case insensitive and matches any part of the file name. Regular expressions are not currently supported.

☐ Date: ... to ...

*Empty fields mean "No Limit" *The date format is mm/dd/yyyy

Timezone:

☒ Modified ☒ Accessed ☒ Created ☒ Changed

☐ Size:

☐ Known Status:

☒ Unknown ☒ Known (NSRL or other) ☒ Notable

☐ MIME Type:

application/activemessage

application/andrew-inset

application/applefile

application/applixware

application/atom+xml

*Note: Multiple MIME types can be selected

☐ Data Source:

LogicalFileSet1

*Note: Multiple data sources can be selected

☐ MD5:

02. Activity: Mobile Evidence Analysis

- In this activity, you will analyze evidence and create a spreadsheet that profiles the details of Peter's iPhone.
- This will serve as your preliminary documentation for the remaining activities.
- For this activity, you will practice your research skills using Autopsy's File Search by Attributes and Keyword Search functions.
- Refer to the following file for information on the main databases and files covered in the demonstration:
 - [iPhone Forensic Image: Important Databases and Files](#)
- Document your findings in the following worksheet:

- [iPhone Details Worksheet](#)
- You will need to make a copy in order to edit it.

If you wish, you may work in groups of two or three.

03. Activity Review: Mobile Evidence Analysis

04. Break

05. Tagging Evidence

We'll continue our investigation by using Autopsy to tag evidence.

- Evidence tagging is the process of bookmarking evidence to keep critical details organized and easily accessible.
- We'll be tagging evidence throughout our investigation to help categorize and organize all digital evidence pertinent to the Pure Gold Credit Union case.

Autopsy includes a built-in feature to tag evidence, which allows investigators to easily locate evidence.

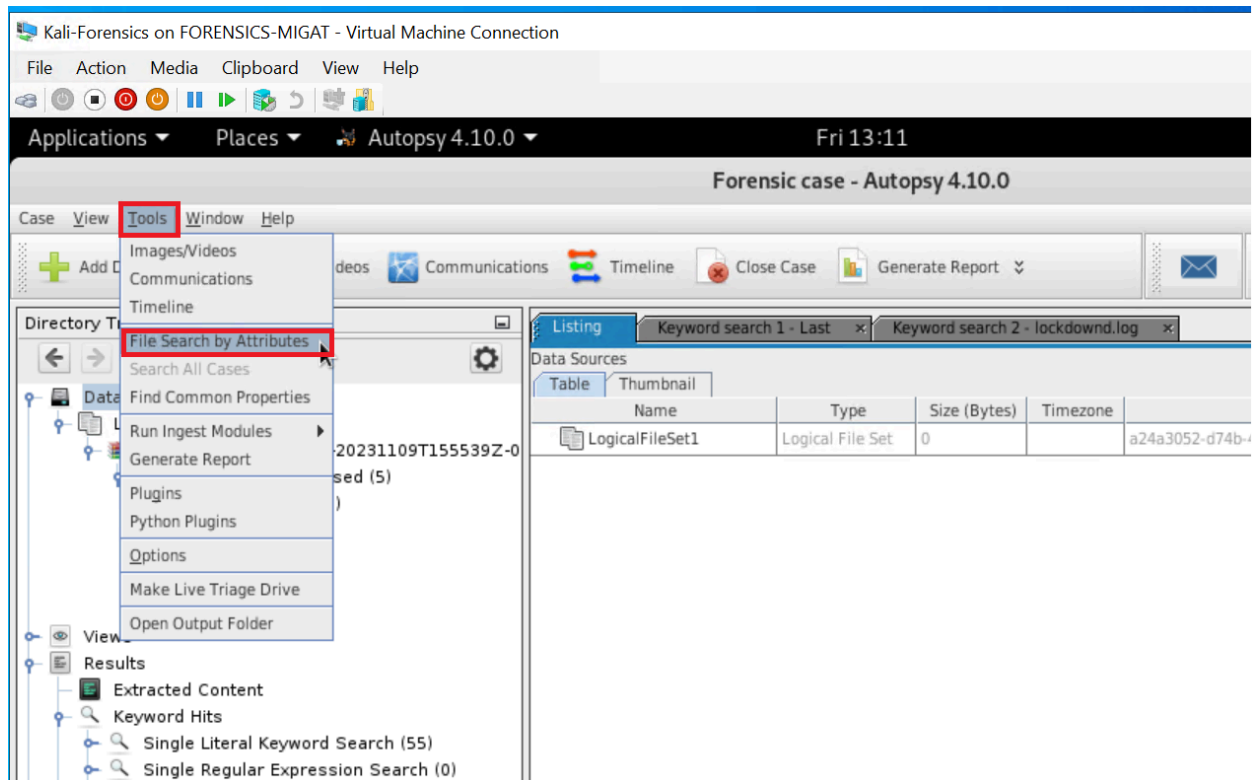
- Autopsy contains a set of predefined tags that include:
 - Follow up
 - Notable item
 - Child exploitation
 - Uncategorized
 - Non-pertinent
- Custom tags can be created and added to the tag database.
 - This allows forensic investigators to categorize and organize their evidence image file database according to specific case criteria.
 - This allows team members to continue working on a file that was previously worked on by another investigator.

Demonstration: Evidence Tagging

Now, we'll walk through how to tag and bookmark the SMS database. The SMS database contains sent and received iPhone messages.

First, we must locate the `sms.db` file in the iPhone image using Files Search by Attributes.

- Click on Tools on the top menu bar.
- Select Files Search by Attributes.



- Check the Name box and enter `sms.db`. Then press Enter or click Search.

File Search by Attributes

☒ Name: sms.db

*Note: Name match is case insensitive and matches any part of the file name. Regular expressions are not currently supported.

Date: ... to ...

*Empty fields mean "No Limit" *The date format is mm/dd/yyyy

Timezone: (GMT-12:00) Etc/GMT+12

☒ Modified ☒ Accessed ☒ Created ☒ Changed

☐ Size: equal to Byte(s)

☐ MIME Type:

application/activemessage

application/andrew-inset

application/applefile

application/applixware

application/atom+xml

*Note: Multiple MIME types can be selected

☐ MD5:

☐ Known Status:

☒ Unknown ☒ Known (NSRL or other) ☒ Notable

☐ Data Source:

LogicalFileSet1

*Note: Multiple data sources can be selected

Search

- Click the sms.db entry in the Listings pane to highlight it.
- Right-click and select Add Tag > Bookmark.

Forensic case - Autopsy 4.10.0

Case View Tools Window Help

+ Add Data Source
 📁 Images/Videos
 ✉ Communications
 📅 Timeline
 🔒 Close Case
 📄 Generate Report

Directory Tree

- Data Sources
 - LogicalFileSet1 (1)
 - Peter's iPhone Parsed-20231109T155539Z-0
 - Peter's iPhone Parsed (5)
 - AddressBook (1)
 - Call History (1)
 - Mail (16)
 - SMS (2)
 - Voicemail (3)
 - Views
 - Results
 - Extracted Content
 - Keyword Hits
 - Single Literal Keyword Search (55)
 - Single Regular Expression Search (0)
 - URLs (206)
 - IP Addresses (28)
 - Phone Numbers (8)

Listing Keyword search 1 - Last Keyword search 2 - lockdown.log **File Search Results 1**

Filename Search Results: 1 Result

Table	Thumbnail	Name	S	C	O	Location	Modified Time	Change Time
		sms.db				LogicalFileSet1; Peter's iPhone Parsed-20231109T155539Z-0	2023-11-08 04:16:16 EST	0000-00-00 00:00:00

Properties

View File in Directory

View in New Window

Open in External Viewer

View File in Timeline...

Extract File(s)

Add File Tag Bookmark Ctrl-B

Remove File Tag

Add File to Hash Set

CAT-0: Uncategorized

CAT-1: Child Exploitation (Illegal) (Notable)

CAT-2: Child Exploitation (Non-Illegal/Age Difficult) (Notable)

CAT-3: CGI/Animation (Child Exploitive) (Notable)

CAT-4: Exemplar/Comparison (Internal Use Only)

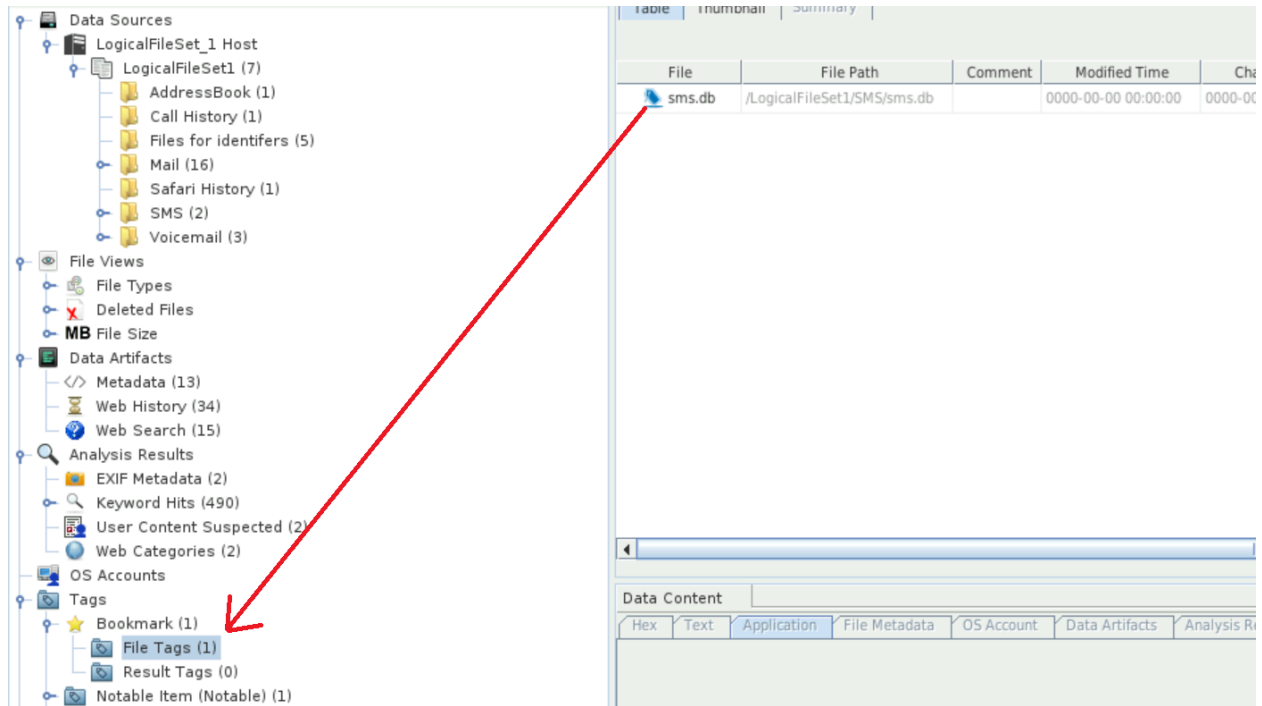
CAT-5: Non-pertinent

Follow Up

Notable Item (Notable)

Tag and Comment...

New Tag...



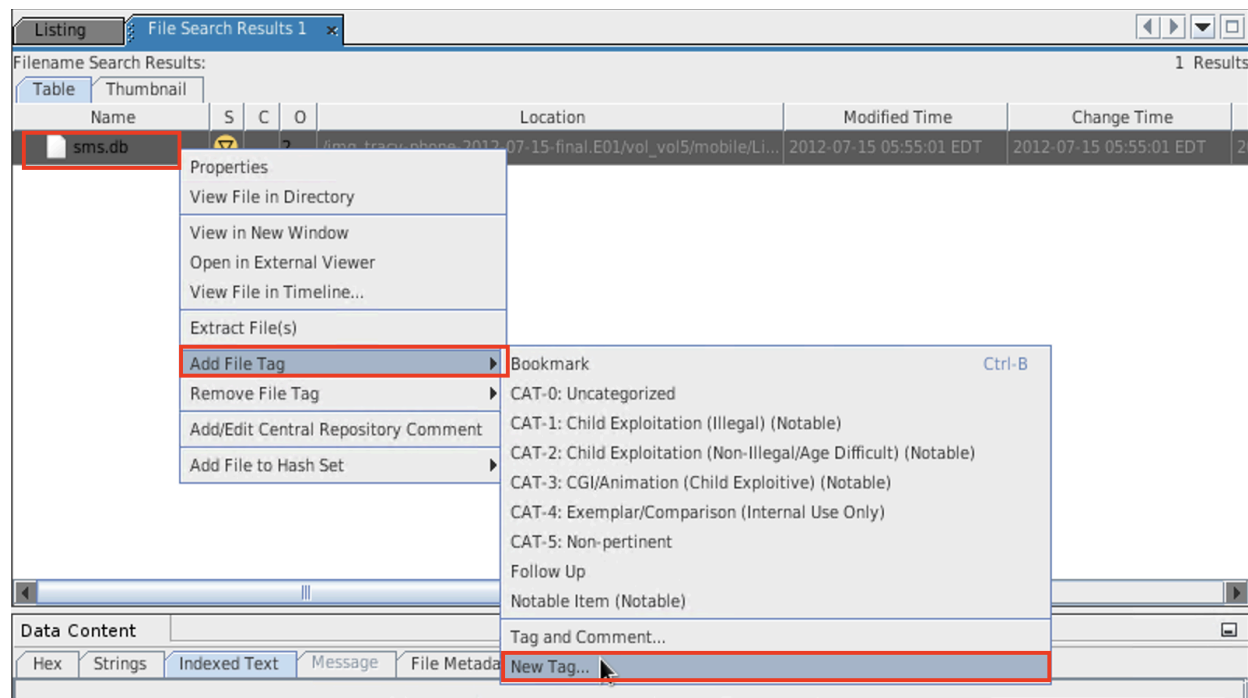
The bookmark entry can be found in the Directory Tree under Tags > Bookmark > File Tags.

Creating Custom Tags

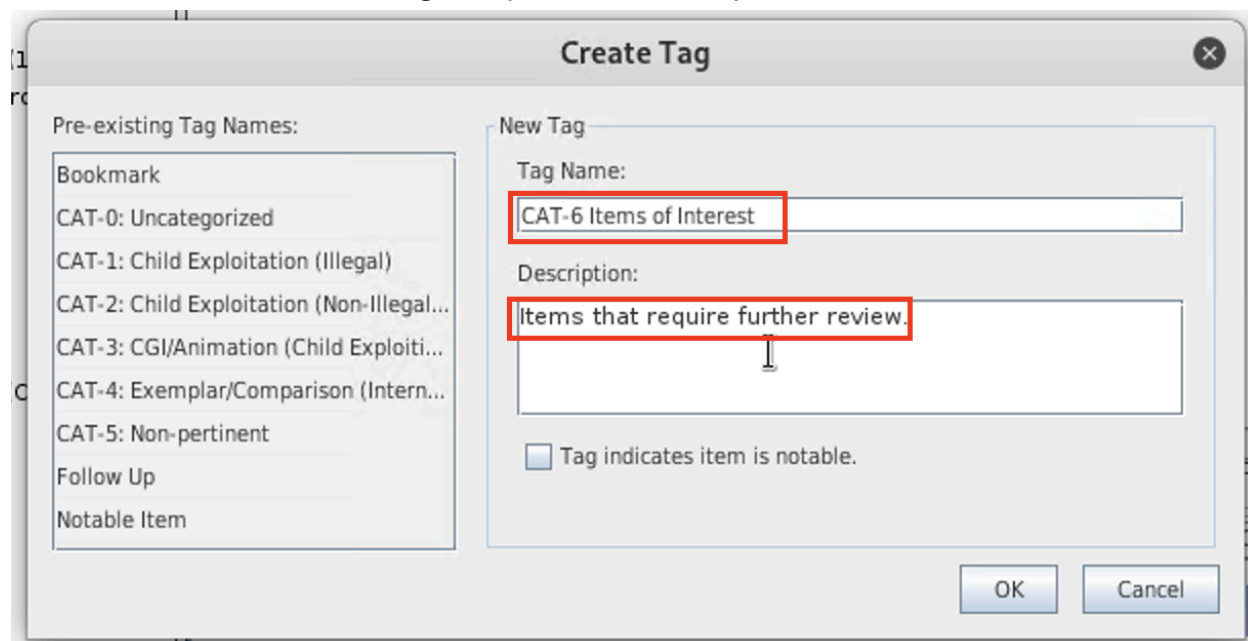
Organizations use various tagging schemes depending on the case they are investigating. Therefore, having the ability to create custom tags allows forensic investigators to categorize evidence according to specific criteria.

- Next, we'll preview how to create a new tag in the database.

- Right-click on `sms.db` and select Add File Tag > New Tag.



- The Create Tag window is displayed.
- Enter the name of the new tag and provide a description. Click OK.



- After you add your custom tag to the `sms.db` file, you will see the new tag entry in the directory tree under Tags.
- The next time you go to select a tag, you will see your custom tag listed as a selection.

06. Activity: Tagging Evidence

- In this activity, you will tag the major databases and files in the iPhone image file.
- You will create new tags for relevant evidence, such as mail, address book, and SMS.
- Be sure to tag as much relevant information as possible. It will be used in later activities as you build your case.

07. Activity Review: Tagging Evidence Activity

08. Extracting Data for Offline Analysis

Although Autopsy facilitates viewing information in the application, investigators may want to extract individual files or entire directory trees for offline viewing in other applications. This is because:

- Not all data types can be rendered in Autopsy.
- Other tools can analyze video, photo, and audio files in more depth.
 - For example, photos may need to be enlarged, audio may need voice recognition, and video may need to be enhanced for facial recognition.
- Database information may easily be transferred to spreadsheets or Word documents where it can be manipulated for reports.

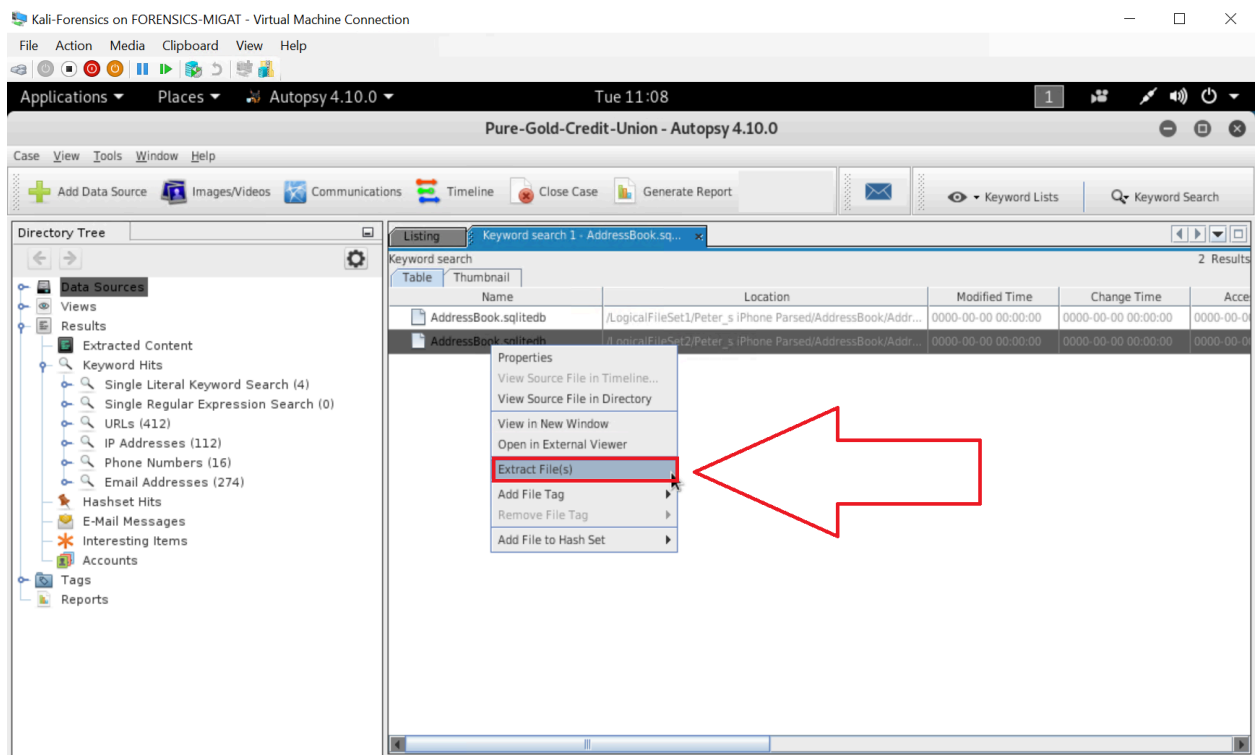
In this section, we will cover how to extract files and entire directories for offline viewing, where data can be parsed for specific information.

- Offline viewing refers to viewing files outside of the main program – in this case, Autopsy.
- Remember that the iPhone image contains files and directories that can be viewed or rendered with other applications on the Linux OS.
- When using the Extract File(s) feature to extract a single file, an entire directory, or the entire image, it will be exported as a Linux directory.

Demonstration: Single File Export

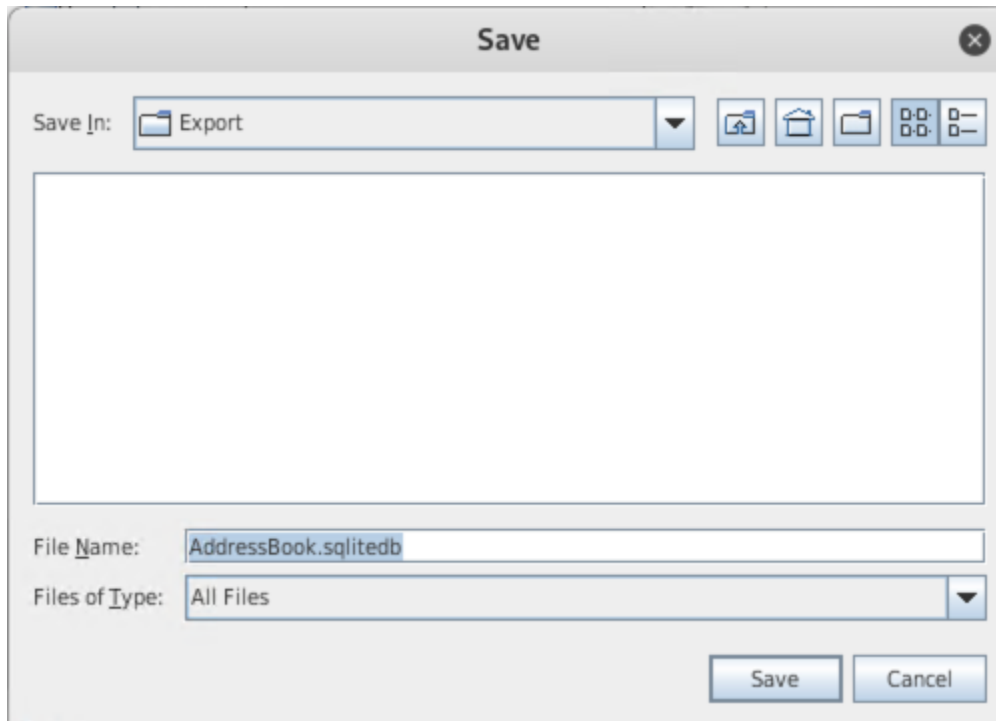
Launch Autopsy and load the case file.

- Perform a Keyword Search and search for the AddressBook.sqlite file.
- In the Listing window, select the AddressBook.sqlite database, then right-click and select Extract Files.



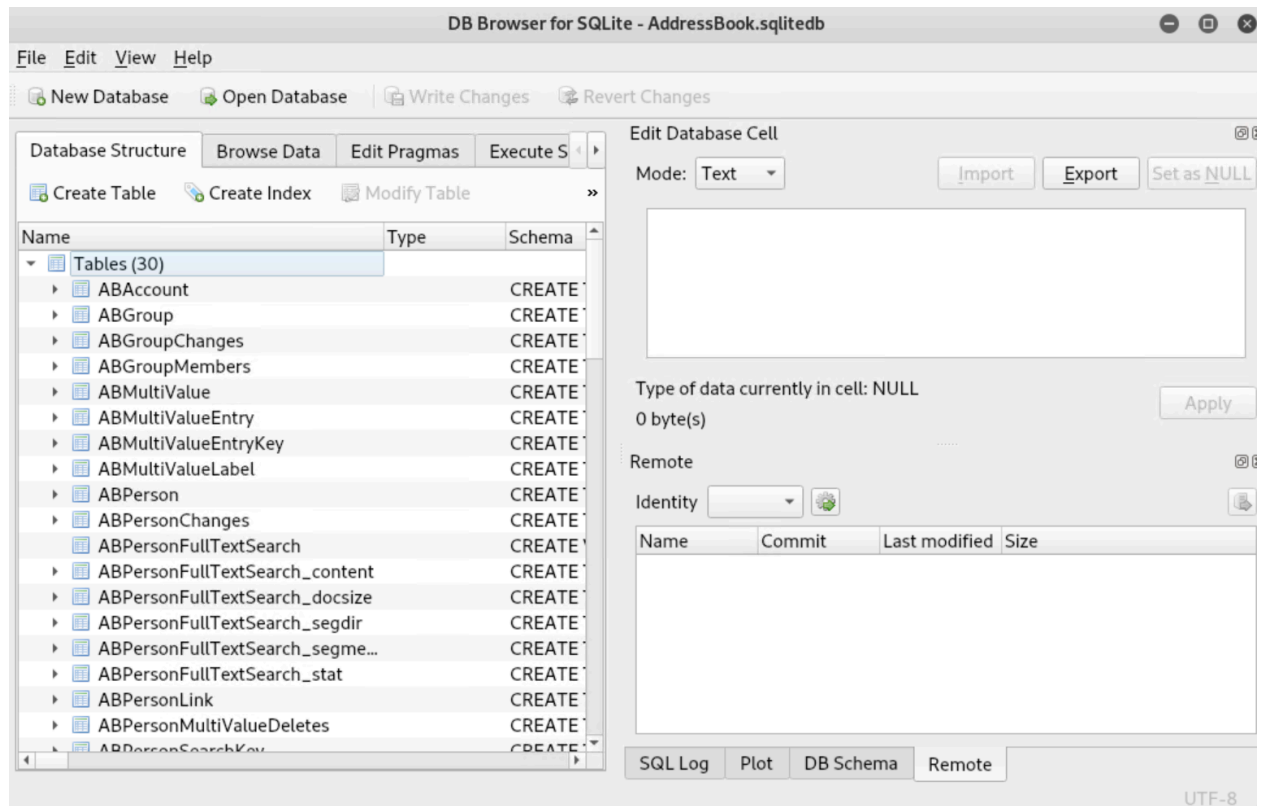
- By default, files are placed in the Export directory.

- Accept the default settings and click Save.

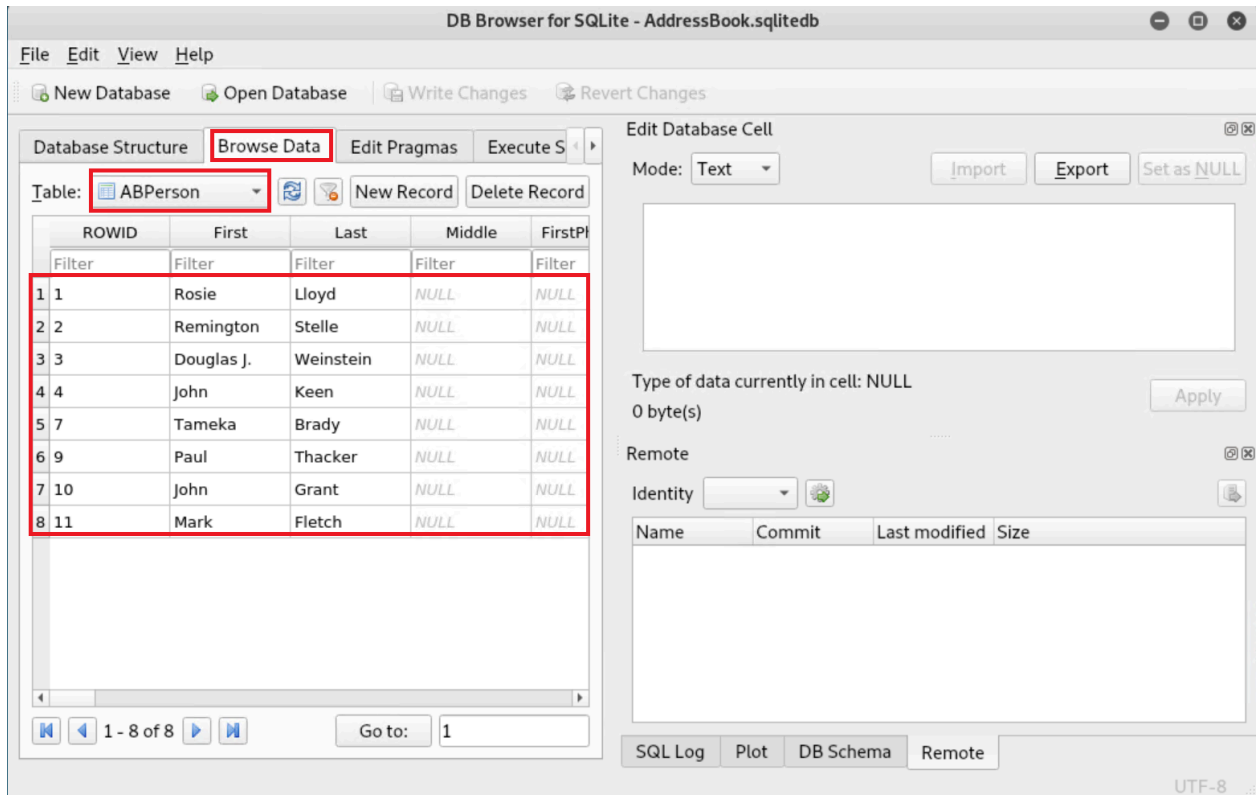


- The Export directory is located in the /home/kali/caseFiles/Pure-Gold-Credit-Union directory.
- Open a terminal and navigate to the export directory.
 - Run `cd /home/kali/caseFiles/Pure-Gold-Credit-Union/Export/`.
- The .db database files require a database reader program such as SQLite DB Browser (pre-installed in Kali) that can view these files types.
 - Run `sqlitebrowser AddressBook.sqlitedb`.
 - Note: If it asks you to install, proceed with the recommended installation.

- We can see the database schema of AddressBook.sqlite3 using SQLite DB Browser.



- We can see the contacts in the ABRecent table by going to Browse Data > ABPerson.



- This provides valuable information, such as the contact's name and phone. However, fields like date and property are in formats designed for the iPhone's operating system.

09. Activity: Extracting Data for Offline Analysis

- In this activity, you will create export files for offline examination.
- The investigative team will use your file exports to identify and confirm Peter's web searches.
- [Activity File: Extracting Data for Offline Analysis](#)

10. Activity Review: Extracting Data for Offline Analysis

11. Wrap-Up

Note the key takeaways from today's class:

- As a forensic investigator, you will work as part of a large collaborative effort.
- As such, it's critical to understand how to use tools such as Autopsy and how to export data so other team members can perform offline analysis of evidence.
- Tagging evidence helps categorize and label evidence that has already been screened.
- This eliminates double work and helps other investigative team members continue your work if you become unavailable.

In the next class, we will continue our investigation by analyzing email messages, SMS messages, and more to tie Peter to the case.